GOBIKATHEESH KARUNAKARAN

Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 3+ years of hands-on experience building and deploying production web applications in Python, JavaScript, TypeScript, React.js, and Node.js/Express.js, with additional backend experience in Java. Skilled in RESTful API design, microservices architecture, database schema design, and SQL/NoSQL systems (MySQL, PostgreSQL, MongoDB, MariaDB). Experienced with containerization (Docker, Kubernetes), AWS cloud infrastructure (EC2, Lambda, S3, RDS), and CI/CD automation with GitHub Actions. Complements this engineering foundation with applied AI coursework through Google Research and IEEE programs (machine learning, deep learning, and real-world AI applications). Delivered production systems across healthcare, university administration, POS, and IoT domains, with a track record of translating business requirements into scalable, reliable software.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design

Backend: Node.js, NestJS, Express.js, RESTful API Design, Microservices Architecture, Laravel

Mobile Development: React Native

Databases: MySQL, PostgreSQL, MongoDB, MariaDB, Firebase, Database Schema Design

Cloud & DevOps: AWS (EC2, Lambda, S3, RDS), Docker, Kubernetes, GitHub Actions (CI/CD)

AI/ML Exposure: Applied AI & Machine Learning coursework (Google Research, IEEE); familiar with core ML/AI concepts

Tools & Version Control: Git, GitHub, Linux, VS Code, Terraform, Jest

Methodologies: Agile/Scrum, Code Review, Technical Documentation, SDLC collaboration

PROFESSIONAL EXPERIENCE

Programmer cum System Analyst *University of Jaffna | May 2025 – Present*

- Designed and developed MyUOJ, a full-stack student enrollment system, using React.js, Node.js, and Express.js, improving system workflows for administrative staff and students.
- Architected microservices and designed MariaDB database schemas to support scalable, reliable data management.
- Built and maintained CI/CD pipelines with GitHub Actions and containerized deployments using Docker, reducing manual release effort.
- Automated student enrollment processes with role-based access control, cutting down manual administrative operations.
- Collaborated cross-functionally with academic and technical stakeholders throughout the SDLC to deliver user-focused features.

Software Engineer *Velxgen | Mar 2024 – Present*

- Developed VELX-POS, a full-stack point-of-sale system for sales and inventory management, using React.js and Laravel (PHP).
- Designed and implemented RESTful APIs and a modular backend architecture to support scalable system growth.
- Built responsive, real-time interfaces for billing and product management used in live retail environments.
- Implemented secure authentication and role-based access control for multi-user, multi-role environments.
- Partnered with stakeholders to translate business requirements into functional, production-ready software features.

Intern Software Engineer *Velxgen | Jun 2023 – Mar 2024*

- Contributed to full-stack feature development on VELX-POS, building React.js frontend components and Laravel backend CRUD endpoints under senior engineer mentorship.
- Built product listing UI and connected frontend views to REST API responses, gaining hands-on exposure to end-to-end data flow in a production codebase.
- Wrote and tested database queries for product and inventory tables, strengthening SQL fundamentals through iterative code review.
- Participated in Agile sprint planning, stand-ups, and retrospectives, gaining practical exposure to professional software delivery.

KEY PROJECTS

Gynecology Ward Management System (Final Year Project) — React.js, Node.js, Express.js, MySQL, Tailwind CSS, JWT | Nov 2023 – Oct 2024

- Built a full-stack hospital management system with secure JWT authentication and role-based access.
- Designed RESTful APIs for managing patient records, appointments, and clinical operations.
- Deployed a responsive frontend that improved hospital workflow efficiency for ward staff.

Appointment Booking System — React.js, Node.js, Express.js, MySQL, Tailwind CSS, JWT | Feb 2025 – Apr 2025

- Built a real-time booking system with role-based access and RESTful API integration.
- Integrated a MySQL database and optimized API endpoints for performance and scalability.

Complaint Management System — HTML, CSS, PHP, JavaScript | May 2023 – Aug 2023

- Developed the frontend interface for a university-wide complaint management system as frontend developer.
- Collaborated with the backend team to integrate APIs and ensure seamless data flow.
- Built automated workflows to efficiently route and handle student and staff grievances.

Personal Portfolio Website — React.js, Tailwind CSS | Aug 2024 – Nov 2024

- Designed and built a responsive, modern portfolio site showcasing projects and technical expertise.

Room Temperature Controller (IoT) — Tinkercad, Arduino IDE, C | Aug 2023 – Sept 2023

- Built an automated IoT system for temperature regulation using embedded sensors.

EDUCATION

Bachelor of Science in Computer Science *University of Jaffna | Oct 2021 – Dec 2024*

- Graduated: February 2026

G.C.E. Advanced Level – Physical Science (B, 2C) | G.C.E. Ordinary Level (2A, 4B, 3C)

Km/Wesley High School, Kalmunai | 2005 – 2019

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- GeSSoMI Summer School – AI & Machine Learning — Google Research, University of Melbourne, University of Jaffna (2024)
- AI Horizons: Real-World AI Applications — Google Research Award, University of Jaffna (2024)
- Google exploreCSR Symposium – AI Research — Department of Computer Science, University of Jaffna (2024)
- Mastering Deep Learning (Hands-On) — IEEE Computer Society (2024)
- IEEE Summer School on Computational Intelligence — IEEE Computational Intelligence Society (2023)
- Sri Lanka Student Workshop on Computer Science (SL-SWCS'23) — University of Jaffna (2023)
- Research Methods in Computer Science: Theory and Ethics — University of Jaffna (2023)

REFERENCES

Dr. E. Y. A. Charles, Senior Lecturer, Department of Computer Science, University of Jaffna | charles.ey@univ.jfn.ac.lk | +94774696795

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